

Area scale factor



- 1 a 100 b $\frac{9}{100}$ c $\frac{1}{9}$
- 2 a 2 b $\frac{1}{2}$ c 6 d 24 units²
e 4 units²
- 3 a $x = 36 \text{ cm}^2$ b $x = 504 \text{ cm}^2$
- 4 a $\frac{3}{7}$ b $\frac{9}{49}$
- 5 a 2 cm b 36 cm² c 9
- 6 270 cm²
- 7 324 cm²
- 8 600 cm²
- 9 The area will increase by a factor of 9.
- 10 a 21, 30, 48 b 99 cm
- 11 a 128 cm² b 9 cm
- 12 a The value of A will be doubled. b A will increase by a factor of 4.

13

a A will increase by a factor of 9.b A will decrease by a factor of 16.

Volume scale factor

14

a 2

b 8

15

a 4

b 64

c $54\,528\text{ cm}^3$

Surface area scale factor

16

a 5

b 25

c 177 cm^2

17

a $\frac{8}{7}$ b $\frac{512}{343}$

18

a 4

b 2

c 26 mm

19

a $\frac{11}{5}$ b $\frac{121}{25}$

20

a 1.4

b 1.96

c 2.744

Applications

21 4 cm

22 15 cm

23

a 6

b 36

c $14\sqrt{2}$



24

a 60 cm^2

b 240 cm^2

c 2

d 4

25 3

26

a 2 : 1

b 8 : 1

c 8

27

a 5

b 25

c 9375 cm^2

28

a 9

b 27

29

a 2 : 1

b 4 : 1

c 8 : 1

d 16 : 1

30

a 4

b 16

c 24 L

31

a 1 : 400

b 1 : 8000

c 7.2 L

d 6 mL

32

a 1 : 20

b 1 : 400

c 1 : 8000